

Student No.

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Student Name:

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Teacher Name:

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2020 TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Approved calculators may be used
- A reference sheet is provided
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
70

Section I – 10 marks (pages 2 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 9 – 14)

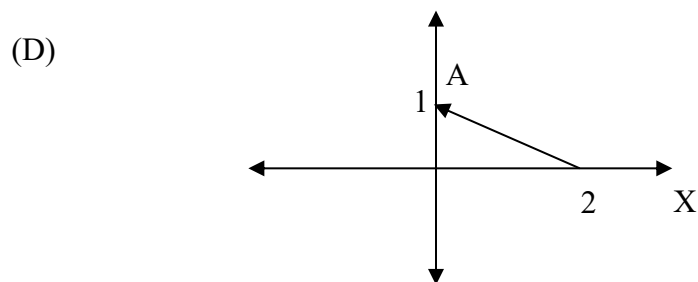
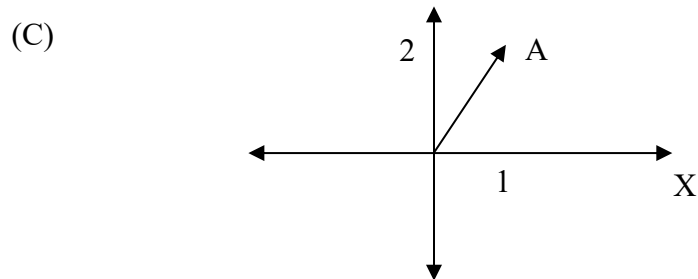
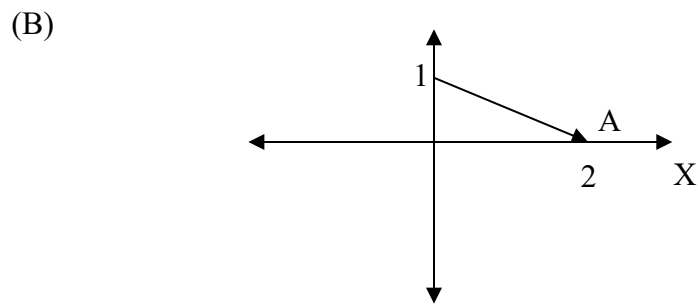
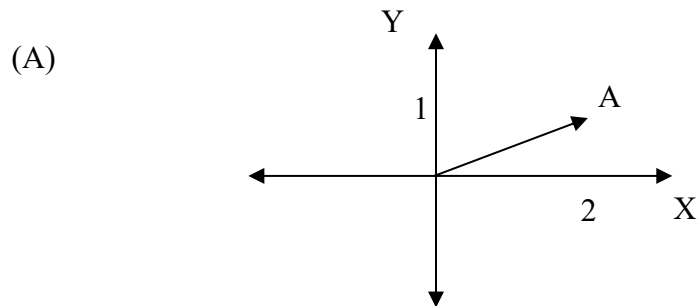
- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Section I**10 marks****Attempt Questions 1–10****Allow about 15 minutes for this section**

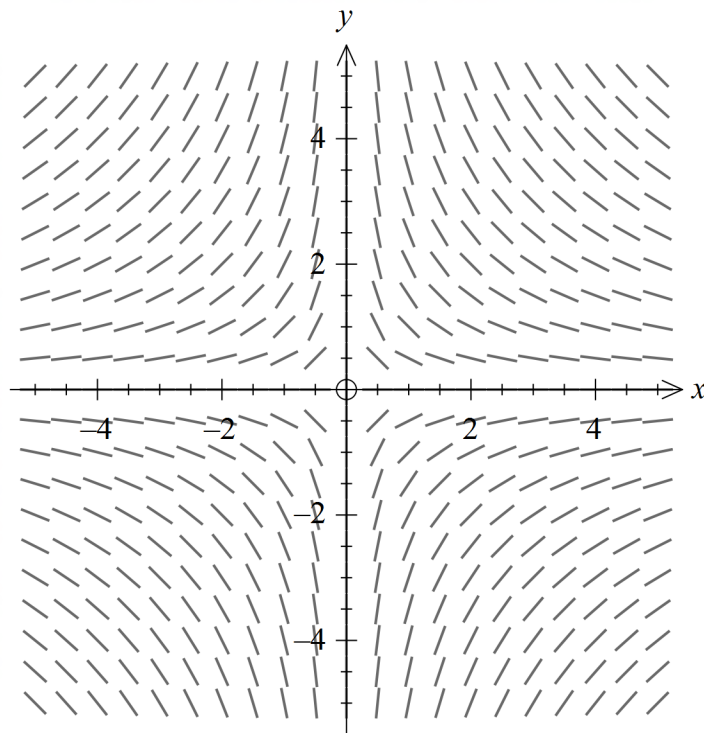
Use the multiple-choice answer sheet for Questions 1 – 10.

1. What is the derivative of $\cos^{-1}\left(\frac{x}{5}\right)$?
- (A) $-\frac{1}{\sqrt{25-x^2}}$
- (B) $\frac{1}{\sqrt{25-x^2}}$
- (C) $-\frac{1}{\sqrt{5-x^2}}$
- (D) $\frac{1}{\sqrt{5-x^2}}$
2. What is the rate of change of the volume of a cube, when the side length of the cube is 4 cm and the surface area is increasing at $2 \text{ cm}^2\text{s}^{-1}$?
- (A) $1 \text{ cm}^3\text{s}^{-1}$
- (B) $2 \text{ cm}^3\text{s}^{-1}$
- (C) $8 \text{ cm}^3\text{s}^{-1}$
- (D) $18 \text{ cm}^3\text{s}^{-1}$
3. An examination consists of 30 multiple-choice questions, each question having five possible answers. A student guesses the answer to every question. Let X be the number of correct answers. What is $\text{P}(X \geq 10)$?
- (A) 5
- (B) 6
- (C) 9
- (D) 15

4. The position vector $\vec{OA} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is best represented by



5. The slope field for a first order differential equation is shown.

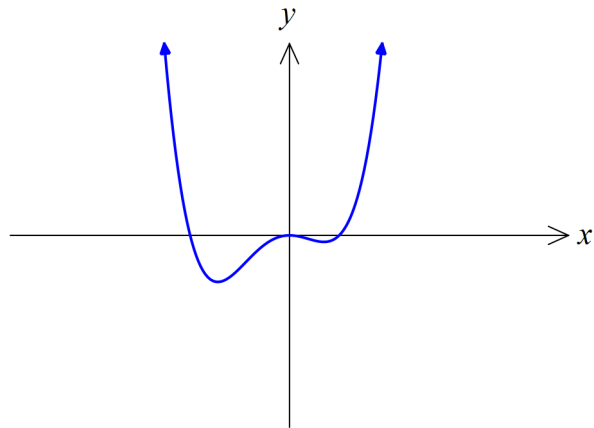


What could be the differential equation represented?

- (A) $\frac{dy}{dx} = -\frac{x}{y}$
- (B) $\frac{dy}{dx} = \frac{x}{y}$
- (C) $\frac{dy}{dx} = -\frac{y}{x}$
- (D) $\frac{dy}{dx} = \frac{y}{x}$
6. Which of the following is true for the vector $\vec{OB} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$

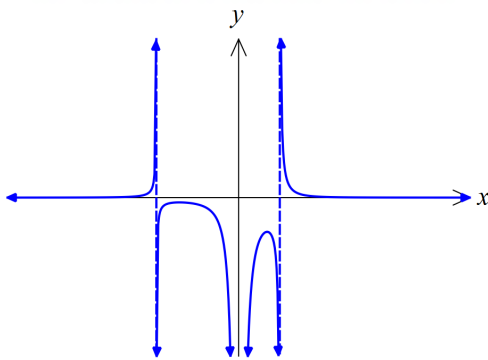
- (A) magnitude = 5 and direction is 53°
- (B) magnitude = 25 and direction is 37°
- (C) magnitude = 5 and direction is 37°
- (D) magnitude = 7 and direction is 37°

7 The diagram shows the graph of $f(x)$.

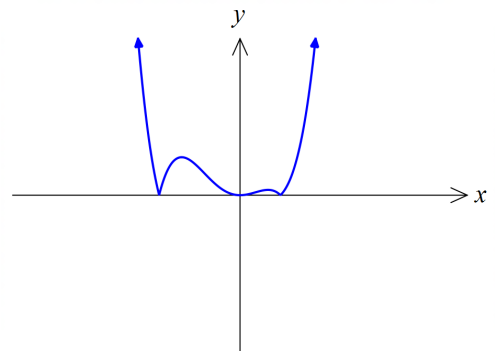


Which graph best represents the function $y = |f(x)|$?

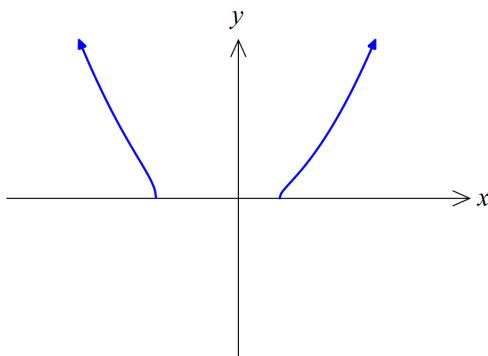
(A)



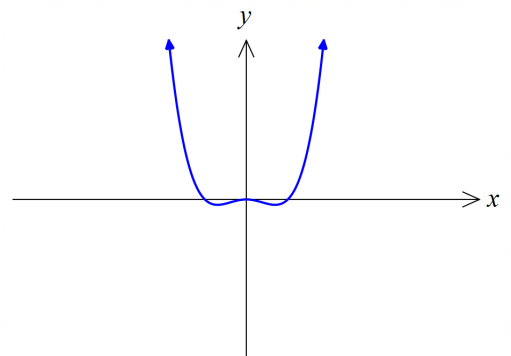
(B)



(C)



(D)



8. Which of the following is **NOT** equal to the area of the region bounded by the curves $y = \sin x$ and $y = \cos x$ between $x = 0$ and $x = \frac{\pi}{4}$?

(A) $-\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos x - \sin x \, dx$

(B) $\int_{-\frac{\pi}{4}}^0 \cos x - \sin x \, dx$

(C) $\int_{-\frac{3\pi}{4}}^{-\frac{\pi}{2}} \cos x - \sin x \, dx$

(D) $\int_{\pi}^{\frac{5\pi}{4}} \cos x - \sin x \, dx$

9. $\cos 3\theta + \sin 3\theta$ in the form $R \cos(3\theta - \alpha)$ and α is in degrees is

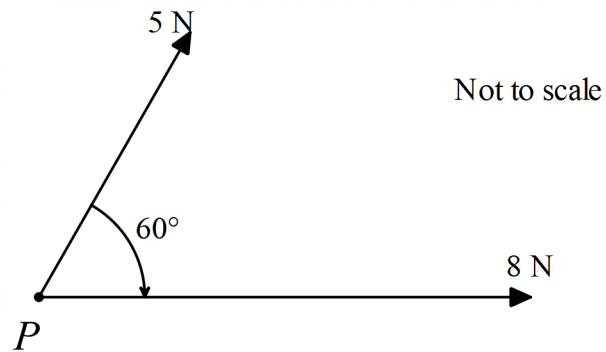
(A) $\sqrt{2}[\cos(3\theta - 45)]$

(B) $\sqrt{2}[\cos(3\theta + 45)]$

(C) $\frac{\sqrt{3}}{2}[\cos(3\theta - 45)]$

(D) $\frac{\sqrt{3}}{2}[\cos(3\theta + 45)]$

10. Forces of magnitude 8 N and 5 N act on a particle P . The angle between the directions of the two forces is 60° as shown in the diagram.



Which of the following is the correct magnitude and direction of the resultant force acting on P ?

- (A) 11.36 N, $22^\circ 25'$ to the horizontal
- (B) 11.36 N, $67^\circ 35'$ to the horizontal
- (C) 12.58 N, $22^\circ 25'$ to the horizontal
- (D) 12.58 N, $67^\circ 35'$ to the horizontal

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Section II**60 marks****Attempt Questions 11 – 14.****Allow about 1 hour and 45 minutes for this section.**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet.

a) Solve $\frac{x+2}{x-5} \geq 0$. 2

b) The polynomial equation $x^3 - 5x^2 + x + 3 = 0$ has roots α, β and γ . Find $\alpha^2 + \beta^2 + \gamma^2$. 2

c) i) Write an expression for $\sin 5x \sin x$ in terms of $\cos 4x$ and $\cos 6x$. 2

ii) Hence, find $\int_0^{\frac{\pi}{4}} \sin 5x \sin x \, dx$. 2

d) Weather records suggest that in the town of Mathsville, an average of 5 days out of 30 in the month of April will be wet. 3

If X represents the number of wet days in April, such that $X \sim \text{Bin}\left(30, \frac{1}{6}\right)$, find

$P(X \geq 2)$.

e) The column vector notation of four vectors is shown below.

$P = \begin{pmatrix} -8 \\ -8 \end{pmatrix}$ $Q = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$ $R = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$ $S = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$. Find

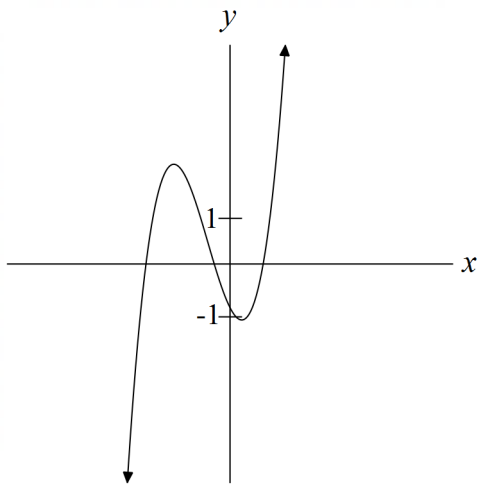
i) $\overrightarrow{PQ}$ 1

ii) $-\overrightarrow{PQ} - \overrightarrow{RS}$ 1

Question 11 continues on next page

f) The graph of $f(x)$ is shown in the diagram.

2



Sketch the graph of $y^2 = f(x)$.

End of Question 11

Question 12 (13 marks) Start a new writing booklet.

- a) A game is played in which players throw netballs, aiming to get them through the hoop.

The players are given two options in how they play the game.

Option 1: Players take 2 attempts and win a prize if at least one of the balls goes through.

Option 2: Players take 3 attempts and win a prize if at least two of the balls go through.

Karen decides to play.

The probability that she will get a ball through on any attempt is p , where $p > 0$.

- i) Show that the probability that Karen wins if she takes Option 1 is $2p - p^2$. 1
- ii) Show that the probability that Karen wins if she takes Option 2 is $3p^2 - 2p^3$. 1
- iii) Karen has calculated that she is 3 times as likely to win by choosing Option 1 over Option 2. Find the value of p . 2

- b) Consider the statement $P(n)$:

$$2^0 + 2^1 + 2^2 + \dots + 2^{n-1} = 2^n - 1$$

and the following attempted proof of this statement by induction.

Proof:

Assume the statement is true for $n = k$. That is,

$$2^0 + 2^1 + 2^2 + \dots + 2^{k-1} = 2^k - 1 \quad (1)$$

Next, we shall show it is true for $n = k + 1$ by noting that if

$$2^0 + 2^1 + 2^2 + \dots + 2^{k-1} + 2^k = 2^{k+1} - 1$$

is true, then

$$2^0 + 2^1 + 2^2 + \dots + 2^{k-1} + 2^k = 2 \times 2^k - 1$$

$$2^0 + 2^1 + 2^2 + \dots + 2^{k-1} + 2^k = 2^k + 2^k - 1$$

Now subtracting 2^k from both sides of this equation, we have

$$2^0 + 2^1 + 2^2 + \dots + 2^{k-1} = 2^k - 1$$

Which is true by statement (1). Therefore, by the principle of induction, the statement $P(n)$ is true.

- i) Give two reasons why the given proof is incorrect and does not prove $P(n)$. 2
 - ii) Provide a correct complete proof by induction of the statement $P(n)$. 2
- c)
- i) Use the substitution $t = \tan \frac{x}{2}$ to show that $\operatorname{cosec} x + \cot x = \cot \frac{x}{2}$. 2
 - ii) Hence evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x) dx$. Answer is simplest exact form. 3

End of Question 12

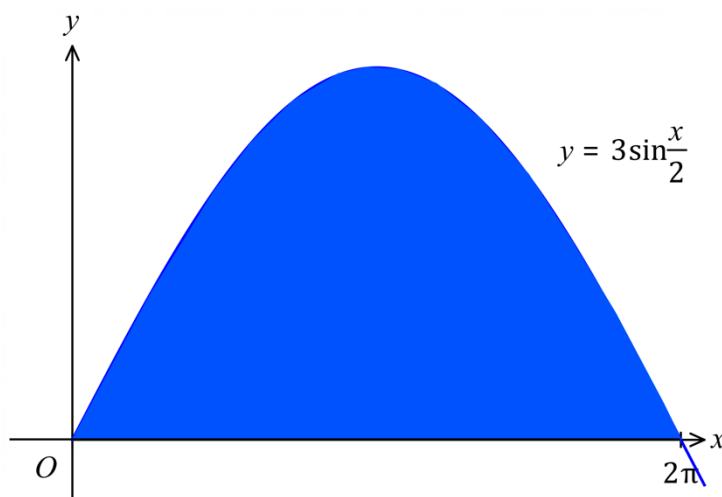
Question 13 (17 marks) Start a new writing booklet.

- a) On a roulette wheel, there are 18 red numbers, 18 black numbers, and one green number. A ball is dropped onto the spinning wheel and lands on one of the numbers randomly. Each result is independent.

A gambler bets that the ball will land on any of the 18 black numbers.

- i) Define the gambler's bet as a Bernoulli random variable X and give its mean and variance. 2
- ii) If the gambler makes the same bet five times, let the random variable Y be the number of times the gambler wins. Describe the distribution of Y , and give its mean and variance. 2
- iii) If the gambler makes the same bet five times, what is the probability he will win more times than he loses? Give your answer correct to three decimal places. 2

b)



Shaded region is enclosed by the curve $y = 3\sin\frac{x}{2}$, $0 \leq x \leq 2\pi$ and the x axis.

- i) Find the area of the shaded region. 1
- ii) This region is rotated through 2π radians about the x -axis. Find the volume of the solid generated. 2

Question 13 continues on next page

c) A tank contains 2,500 litres of water and 25 kg of dissolved salt. Fresh water enters the tank at a rate of 20 litres per minute. The solution is thoroughly mixed at all times and is drained from the tank at a rate of 15 litres per minute.

- i) Using y for the amount of salt in the tank in kilograms (as a function of time), and t for time in minutes, show that the concentration of salt in the tank at time t can be given by 1

$$C = \frac{y}{2500 + 5t}$$

- ii) Explain why the rate of change of salt in the tank can be given by $y' = -15C$. 1

- iii) Find y , the amount of salt in the tank as a function of t . 4

- iv) If the tank has a capacity of 5,000 litres, how much salt is in the tank when it overflows? 2

End of Question 13

Question 14 (15 marks) Start a new writing booklet.

a) For the differential equation $y' = \frac{6}{5x^2 + 4x - 1}$:

i) Show that $y' = \frac{5}{5x - 1} - \frac{1}{x + 1}$. 1

ii) Find the solution to the differential equation; given that when $x = \frac{1}{2}$, $y = 3$. 2

b) A golf ball is hit at a velocity of 110 m/s at an angle θ , to the horizontal.

The position vector $s(t)$, from the starting point, of the ball after t seconds is given by

$$s = 110t \cos\theta \mathbf{i} + (110t \sin\theta - 4.9t^2)\mathbf{j}$$

i) Using gravity of 9.8 ms^{-2} show that the maximum horizontal range of the ball is 2
 $\frac{12100 \sin 2\theta}{9.8}$ metres.

ii) To ensure that the ball lands on the green, it must travel between 400 and 450 metres. What values of θ , correct to the nearest minute, would allow this to happen? 2

iii) The golfer hits the ball directly towards the green with a velocity of 110 m/s. 2

After 3.4 seconds of flight, at a point 8 metres above the ground, the ball hits a low flying TV drone. If it had not hit the drone or any other obstacles, would the ball have landed on the green?

c) Consider the function $y = \cos^{-1}(\sin x)$.

i) Find $\frac{dy}{dx}$. 1

ii) What does your answer to part (a) tell you about stationary points for this function? 1

iii) State the domain and range for this function. 2

iv) Sketch the function over the domain $0 \leq x \leq 2\pi$ 2

End of paper

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Mathematics Extension 1

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Multiple Choice Answer Sheet

Full Name: Solutions

Teacher Name: _____

Completely fill the response oval representing the most correct answer.

1. A ☒ B ☐ C ☐ D ☐
2. A ☐ B ☒ C ☐ D ☐
3. A ☐ B ☒ C ☐ D ☐
4. A ☒ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☒ D ☐
6. A ☐ B ☐ C ☒ D ☐
7. A ☐ B ☒ C ☐ D ☐
8. A ☐ B ☒ C ☐ D ☐
9. A ☒ B ☐ C ☐ D ☐
10. A ☒ B ☐ C ☐ D ☐

Aryan → no name on 2nd booklet

Ruhi → Q13 started on Q12 booklet

Ashna → number her booklets not per question
but total

Stuti, Shardian, Preeti & Ruhi → did not number
& Praneet their booklets 1 of 1



Question 11

Name: _____

Teacher: _____

Question
No.

☐ Year 11

☐ Year 12

☐ Adv

☐ Ext 1

☐ Ext 2

Question 11

$$a) \frac{x+2}{x-5} \geq 0 \quad x \neq 5$$

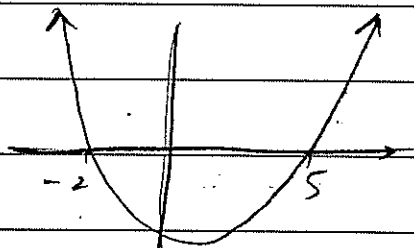
$$\frac{(x+2)(x-5)^2}{x-5} \geq 0(x-5)^2$$

$$(x-5)(x+2) \geq 0$$

$$x^2 - 3x - 10 \geq 0$$

$$(x-5)(x+2) \geq 0$$

$$x \leq -2 \quad x > 5$$



$$b) \alpha + \beta + \gamma = 5$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = 1$$

$$\alpha\beta\gamma = -3$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= (5)^2 - 2(1)$$

$$= 25 - 2 = 23$$

$$c) \sin 5x \sin x = \frac{1}{2} [\cos(5x - x) - \cos(5x + x)]$$

$$= \frac{1}{2} [\cos 4x - \cos 6x]$$

$$(u) \int_0^{\pi/4} \sin 5x \sin x \, dx$$

$$= \frac{1}{2} \int_0^{\pi/4} \cos 4x - \cos 6x$$

$$= \frac{1}{2} \left[\frac{\sin 4x}{4} - \frac{\sin 6x}{6} \right]_0^{\pi/4}$$

$$= \frac{1}{2} \left[\left(\frac{\sin \pi}{4} - \frac{\sin \frac{6\pi}{4}}{6} \right) - \left(\frac{\sin 0}{4} - \frac{\sin 0}{6} \right) \right]$$

$$= \frac{1}{2} \left[\left(0 - \frac{-1}{6} \right) - (0 - 0) \right]$$

$$= \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

$$d) X \sim \text{Bin} \left(30, \frac{1}{6} \right)$$

$$P(X \geq 2) = 1 - P(X < 2)$$

$$= 1 - [P(X=0) + P(X=1)]$$

$$= 1 - \left[{}^{30}C_0 \left(\frac{5}{6} \right)^{30} \left(\frac{1}{6} \right)^0 + {}^{30}C_1 \left(\frac{5}{6} \right)^{29} \left(\frac{1}{6} \right)^1 \right]$$

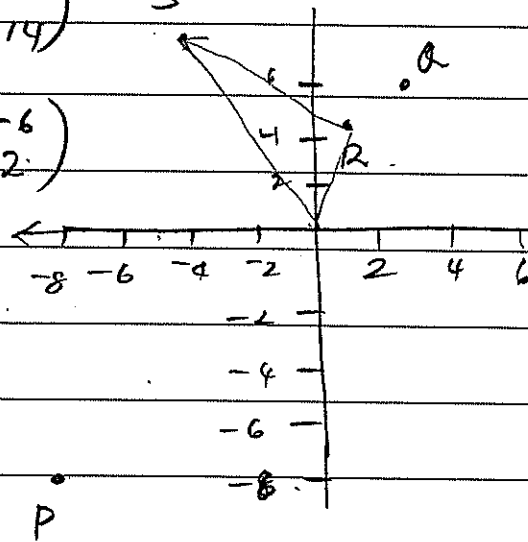
$$= 0.97$$

$$9. (i) \vec{PQ} = \begin{pmatrix} 8+3 \\ 8+6 \end{pmatrix} = \begin{pmatrix} 11 \\ 14 \end{pmatrix}$$

$$(ii) \vec{RS} = \begin{pmatrix} -1-5 \\ -5+7 \end{pmatrix} = \begin{pmatrix} -6 \\ -2 \end{pmatrix}$$

$$(iii) -\vec{PQ} - \vec{RS}$$

$$\begin{pmatrix} -11+6 \\ -14-2 \end{pmatrix} = \begin{pmatrix} -5 \\ -16 \end{pmatrix}$$



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Question 12

Name: _____

Teacher: _____

Question
No.

☐ Year 11

☐ Year 12

☐ Adv

☐ Ext 1

☐ Ext 2

$$\begin{aligned}
 a) (i) P(\text{at least } 1) &= P(X=1) + P(X=2) \\
 &= {}^2C_1 (1-p)^1 (p)^1 + {}^2C_2 (1-p)^0 (p)^2 \\
 &= 2(1-p)(p) + p^2 \\
 &= 2p - 2p^2 + p^2 \\
 &= 2p - p^2
 \end{aligned}$$

$$\begin{aligned}
 (ii) P(\text{at least } 2) &= P(X=2) + P(X=3) \\
 &= {}^3C_2 (1-p)^1 (p)^2 + {}^3C_3 (1-p)^0 (p)^3 \\
 &= 3(1-p)p^2 + p^3 \\
 &= 3p^2 - 3p^3 + p^3 \\
 &= 3p^2 - 2p^3
 \end{aligned}$$

$$(iii) 2p - p^2 = 3(3p^2 - 2p^3)$$

$$2p - p^2 = 9p^2 + 6p^3 = 0$$

$$6p^3 - 10p^2 + 2p = 0$$

$$2p(3p^2 - 5p + 1) = 0$$

$$2p = 0$$

$$p = 0$$

$$\therefore p > 0$$

$$\begin{aligned}
 \therefore p &= \frac{5 \pm \sqrt{(-5)^2 - 4 \times 3 \times 1}}{2 \times 3} & \begin{matrix} 3p & -1 \\ p & = 1 \\ & -1 \end{matrix} \\
 &= \frac{5 \pm \sqrt{13}}{6}
 \end{aligned}$$

Since $0 < p < 1$

$$p = \frac{5 - \sqrt{13}}{6} \approx 0.23$$

b).

(1) ① $n=1$ is not proved.

② $n=k+1$ is simplified on both left & right sides simultaneously, ③ $n=k+1$ used the assumption that $n=1$ is true.

(ii). prove the statement is true for $k=1$

$$\text{LHS } 2^0 = 1 - 1$$

$$\text{RHS } 2^1 - 1 = 1 \quad \text{LHS} = \text{RHS} \text{ hence it's true for } k=1$$

Assume it's true for $n=k$

$$2^0 + 2^1 + 2^2 + \dots + 2^{k-1} = 2^k - 1$$

prove the statement is true for $n=k+1$

$$\text{prove } 2^0 + 2^1 + 2^2 + \dots + 2^k = 2^{k+1} - 1$$

$$2^0 + 2^1 + 2^2 + \dots + 2^{k-1} + 2^k$$

$$2^k - 1 + 2^k$$

$$2 \times 2^k - 1$$

$$2^{k+1} - 1 = \text{RHS}$$

Thus this is true for $n=k+1$ if it's true for $n=k$. By mathematical induction it's true for all integer values of k .

LHS
c) $\csc x + \cot x$

$$= \frac{1+t^2}{2t} + \frac{1-t^2}{2t}$$

$$= \frac{1+t^2+1-t^2}{2t}$$

$$= \frac{2}{2t} = \frac{1}{t} = \frac{1}{\tan \theta/2} = \cot \theta/2$$

$$\int_{\pi/3}^{\pi/2} (\operatorname{cosec} x + \cot x) dx$$

$$= \int_{\pi/3}^{\pi/2} \cot \frac{x}{2}$$

$$= \int_{\pi/3}^{\pi/2} \frac{\cos x/2}{\sin x/2} = 2 \int_{\pi/3}^{\pi/2} \frac{\frac{1}{2} \times \cos x/2}{\sin x/2}$$

$$= 2 \left[\ln \sin x/2 \right]_{\pi/3}^{\pi/2}$$

$$= 2 \left[\left(\ln \sin \pi/4 \right) - \left(\ln \sin \pi/6 \right) \right]$$

$$= 2 \left[\ln \frac{1}{\sqrt{2}} - \ln \frac{1}{2} \right]$$

$$= 2 \left[\ln 2^{-1/2} - \ln 2^{-1} \right]$$

$$= 2 \left[\frac{-1}{2} \ln 2 + \ln 2 \right]$$

$$= 2 \left[\frac{1}{2} \ln 2 \right]$$

$$= \ln 2.$$



Question 13

Name: _____

Teacher: _____

Question
No.

☐ Year 11

☐ Year 12

☐ Adv

☐ Ext 1

☐ Ext 2

Question 13

$$a) X \sim \text{Ber}\left(\frac{18}{37}, 1\right) \quad \left(\frac{18}{37}, \frac{19}{37}\right)$$

$$\mu = \frac{18}{37}$$

$$\sigma^2 = \frac{18}{37} \times \frac{19}{37} = \frac{342}{1369}$$

$$(ii) Y \sim B\left(\frac{18}{37}, 5\right) \quad \left(5, \frac{18}{37}\right)$$

$$\mu = np = 5 \times \frac{18}{37} = \frac{90}{37}$$

$$\sigma^2 = npq = 5 \times \frac{18}{37} \times \frac{19}{37} = \frac{1710}{1369}$$

$$(iii) P(X=3) + P(X=4) + P(X=5)$$

$$= {}^5C_3 \left(\frac{19}{37}\right)^2 \left(\frac{18}{37}\right)^3 + {}^5C_4 \left(\frac{19}{37}\right)^1 \left(\frac{18}{37}\right)^4 + {}^5C_5 \left(\frac{19}{37}\right)^0 \left(\frac{18}{37}\right)^5$$

$$= 0.475$$

$$b). (ii) \text{ Area} = 3 \int_0^{2\pi} \sin x/2 \, dx$$

$$= 3 \left[-\cos x/2 \right]_0^{2\pi}$$

$$= \left[-6 \cos\left(\frac{x}{2}\right) \right]_0^{2\pi}$$

$$= (-6 \cos \pi) - (-6 \cos 0)$$

$$= 6 + 6$$

$$= 12 u^2$$

$$V = \pi \int_0^{2\pi} \left(3 \sin \frac{x}{2} \right)^2 dx$$

$$= \frac{9\pi}{2} \times \int_0^{2\pi} (1 - \cos x) dx$$

$$= \frac{9\pi}{2} \left[x - \sin x \right]_0^{2\pi}$$

$$= \frac{9\pi}{2} \left[(2\pi - \sin 2\pi) - (0 - \sin 0) \right]$$

$$= \frac{9\pi}{2} \left[(2\pi - 0) - (0) \right]$$

$$= \frac{9\pi}{2} \times 2\pi$$

$$= 9\pi^2 u^3$$

$$\cos 2x = 1 - 2 \sin^2 x$$

$$2 \sin^2 x = 1 - \cos 2x$$

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$\sin^2 \frac{x}{2} = \frac{1}{2} (1 - \cos x)$$

c) Amount of liquid in tank = $2500 + 20t - 15t$
 $= 2500 + 5t$

Concentration of salt = $\frac{\text{Amount of salt}}{\text{Total amount of liquid}}$

$$= \frac{y}{2500 + 5t}$$

$$C = \frac{g}{L}$$

(4) Rate of change of salt =
 Use chain rule $(\text{Concentration of salt/litre}) \times \text{Amount of liquid leaving}$
 $\frac{g}{\text{min}} = \frac{g}{\cancel{\text{L}}} \times \frac{\cancel{\text{L}}}{\text{min}}$
 $= C \times -15 \text{ L/min} = -15C$

(41) $y' = -15C$

$$\frac{dy}{dt} = \frac{-15 \times y}{2500 + 5t} \quad \textcircled{1}$$

$$\int \frac{dy}{-15y} = \int \frac{5dt}{2500 + 5t}$$

$$-\frac{1}{15} \ln|y| = \frac{1}{5} \ln|2500 + 5t| + C$$

$$\ln|y| = -\frac{15^3}{5} \times \frac{1}{5} \ln|2500 + 5t| + C \quad \textcircled{2}$$

$$\ln|y| = \ln|2500 + 5t|^{-3} + C$$

y & t are positive

$$\ln y = \ln(2500 + 5t)^{-3} + C$$

$$e^{\ln y} = e^{\ln(2500 + 5t)^{-3} + C}$$

$$y = (2500 + 5t)^{-3} \times e^C$$

$$y = A(2500 + 5t)^{-3}$$

At $t=0$, $y=25$

$$25 = A(2500 + 5 \times 0)^{-3} \quad \textcircled{3}$$

$$A = 25 \times 2500^3$$

$$y = \frac{25 \times 2500^3}{(2500 + 5t)^3}$$

$$\begin{aligned} \text{(iv) Amount of water in tank} &= 2500 + 20t - 15t \\ &= 2500 + 5t \end{aligned}$$

Overflow when

$$5000 = 2500 + 5t$$

$$2500 = 5t$$

$$t = \frac{2500}{5} = 500$$

Amount of salt at $t = 500$

$$y = \frac{25 \times 2500^3}{2500 + 5 \times 500}$$

$$= 3.125 \text{ kg}$$



Question 14

Name: _____

Teacher: _____

Question
No.

☐ Year 11

☐ Year 12

☐ Adv

☐ Ext 1

☐ Ext 2

a) $y' = \frac{6}{5x^2 + 4x - 1}$

(2) $\frac{5}{5x-1} - \frac{1}{x+1} = \frac{5(x+1) - 1(5x-1)}{(5x-1)(x+1)}$

$= \frac{5x+5-5x+1}{(5x-1)(x+1)}$

$= \frac{-6}{(5x-1)(x+1)} = \frac{6}{5x^2 + 4x - 1}$

(4) $\frac{dy}{dx} = \frac{6}{5x^2 + 4x - 1}$

$\int \frac{dy}{dx} = \int \frac{5}{5x-1} - \int \frac{1}{x-1}$

$y = 5 \ln(5x-1) - \ln(x-1) + C$

$y = \ln \frac{5x-1}{x-1} + C$

when $x = \frac{1}{2}$, $y = 2$

$$3 = \ln \left(\frac{\frac{5}{2} - 1}{\left(\frac{1}{2} + 1\right)} \right) + C$$

$$3 = \ln \left(\frac{\frac{3}{2}}{\frac{3}{2}} \right) + C$$

$$3 = \ln 1 + C$$

$$3 = 0 + C$$

$$y = \ln \left(\frac{5x-1}{x+1} \right) + 3$$

$$b) \quad s = 110t \cos \theta \underline{\hat{i}} + (110t \sin \theta - 4.9t^2) \underline{\hat{j}}$$

Max horizontal range when $y = 0$

$$110t \sin \theta - 4.9t^2 = 0$$

$$t(110 \sin \theta - 4.9t) = 0$$

$$t = 0$$

$$110 \sin \theta - 4.9t = 0$$

$$4.9t = 110 \sin \theta$$

$$t = \frac{110 \sin \theta}{4.9}$$

$$\text{when } t = \frac{110 \sin \theta}{4.9}$$

$$x = 110 \times \frac{110 \sin \theta}{4.9} \times \cos \theta$$

$$x = \frac{12100 \sin \theta \cos \theta}{4.9}$$

$$x = \frac{6050 \times 2 \sin \theta \cos \theta}{4.9} = \frac{6050 \times \sin 2\theta}{4.9} = \frac{12100 \sin 2\theta}{9.8}$$

$$(M) \quad 400 < R_{\text{ayl}} < 450$$

$$400 < \frac{12100 \sin 2\theta}{9.8} < 450$$

$$3920 < 12100 \sin 2\theta < 4410$$

$$\frac{196}{605} < \sin 2\theta < \frac{441}{12100}$$

$$18^{\circ}54' < 2\theta < 21^{\circ}22'$$

$$9^{\circ}27' < \theta < 10^{\circ}41'$$

$$(M) \quad \text{At } t = 3.4, \quad y = 8$$

$$y = 110t \sin \theta - 4.9t^2$$

$$8 = 110(3.4) \sin \theta - 4.9(3.4)^2$$

$$8 = 374 \sin \theta - 56.644$$

$$\sin \theta = \frac{8 + 56.644}{374}$$

$$\theta = 9^{\circ}57'$$

θ is between $9^{\circ}27'$ and $10^{\circ}41'$

$\therefore$ the ball would have made it to green.

c) $y = \cos^{-1}(\sin x)$

$$\frac{dy}{dx} = \frac{-\cos x}{\sqrt{1 - \sin^2 x}}$$

$$= \frac{-\cos x}{\sqrt{\cos^2 x}}$$

$$= \frac{-\cos x}{\pm \cos x} = \pm 1$$

(11) Since the gradient is a constant which is not zero the function has no stationary points.

(11) Domain $(-\infty, \infty)$

$$-1 \leq \sin x \leq 1$$

when $\sin x = -1$

$$\cos^{-1}(-1) = \pi$$

when $\sin x = +1$

$$\cos^{-1}(1) = 0$$

$\therefore$ Range $[0, \pi]$

